Faculty of Computer Science

Data Science

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IMDB Review Sentiment Analysis

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Subject area

This project explores classical **NLP pipelines** for predicting the sentiment of IMDB movie reviews. The goal is to classify reviews as positive or negative based on their text. The dataset "IMDB Dataset of 50K Movie Reviews" is **balanced** for sentiment (positive/negative) and contains 5000 reviews (a **subsample** was chosen).

Pipeline steps

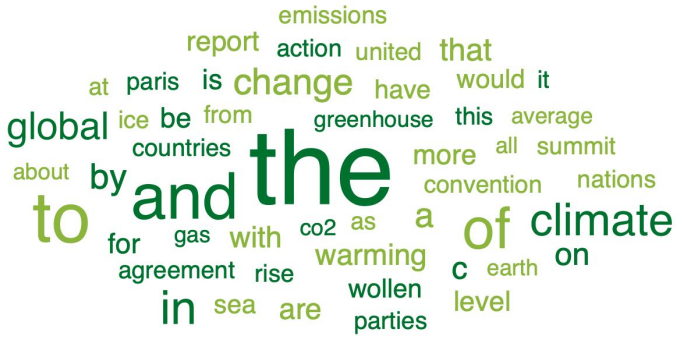
1. **Preprocessing**:
   Tokenization (regex or NLTK), lowercasing, stopword removal, stemming/lemmatization
2. **Feature Extraction**:
   TF-IDF vectorization (unigrams/bigrams).
3. **Classification**:
   Logistic Regression and Multi-Layer Perceptron (MLP)
4. **Evaluation**:
   5-fold stratified cross-validation, reporting mean accuracy, F1, ROC-AUC, and PR- AUC

Preprocessing

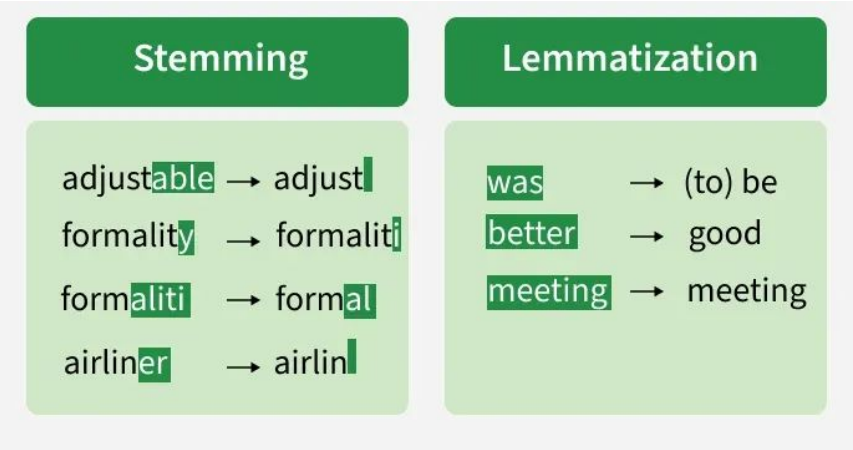
Multiple preprocessing **configurations** were tested, including:

- Regex vs. NLTK tokenization

- With/without stopword removal



- With/without stemming or lemmatization



Experiments results

Table 1: Logistic Regression performance metrics across preprocessing configurations

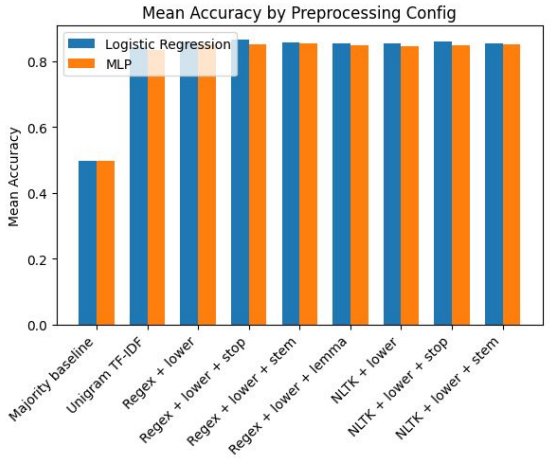
Configuration	Accuracy	F1 Score	ROC-AUC	PR-AUC
Majority baseline	0.4988 ± 0.0000	0.3328 ± 0.0000	0.5000 ± 0.0000	0.5013 ± 0.0000
Unigram TF-IDF	0.8428 ± 0.0099	0.8428 ± 0.0099	0.9215 ± 0.0102	0.9215 ± 0.0102
Regex + lower	0.8560 ± 0.0195	0.8560 ± 0.0195	0.9319 ± 0.0115	0.9321 ± 0.0110
Regex + lower + stop	$\mathbf{0.8646 \pm 0.0120}$	$\mathbf{0.8645 \pm 0.0120}$	$\mathbf{0.9375 \pm 0.0096}$	0.9363 ± 0.0098
Regex + lower + stem	0.8578 ± 0.0152	0.8578 ± 0.0152	0.9365 ± 0.0112	$\mathbf{0.9373 \pm 0.0102}$
Regex + lower + lemma	0.8544 ± 0.0145	0.8544 ± 0.0145	0.9320 ± 0.0114	0.9324 ± 0.0110
NLTK + lower	0.8548 ± 0.0197	0.8548 ± 0.0197	0.9278 ± 0.0122	0.9280 ± 0.0125
NLTK + lower + stop	0.8594 ± 0.0142	0.8593 ± 0.0142	0.9321 ± 0.0102	0.9304 ± 0.0115
NLTK + lower + stem	0.8552 ± 0.0124	0.8552 ± 0.0123	0.9324 ± 0.0115	0.9333 ± 0.0114

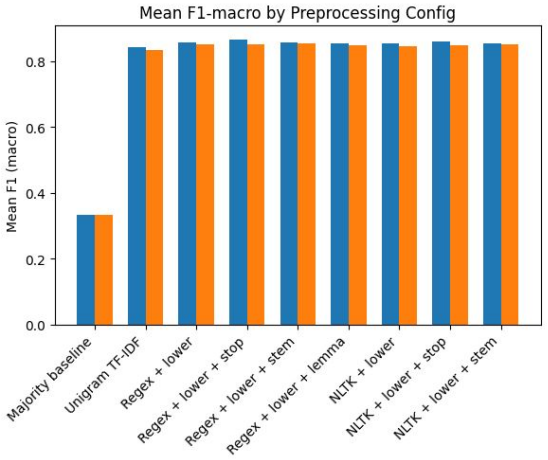
Experiments results

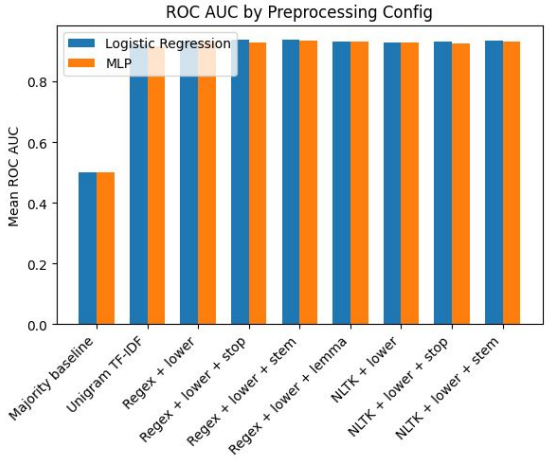
Table 2: Multi-Layer Perceptron (MLP) performance metrics across preprocessing configurations

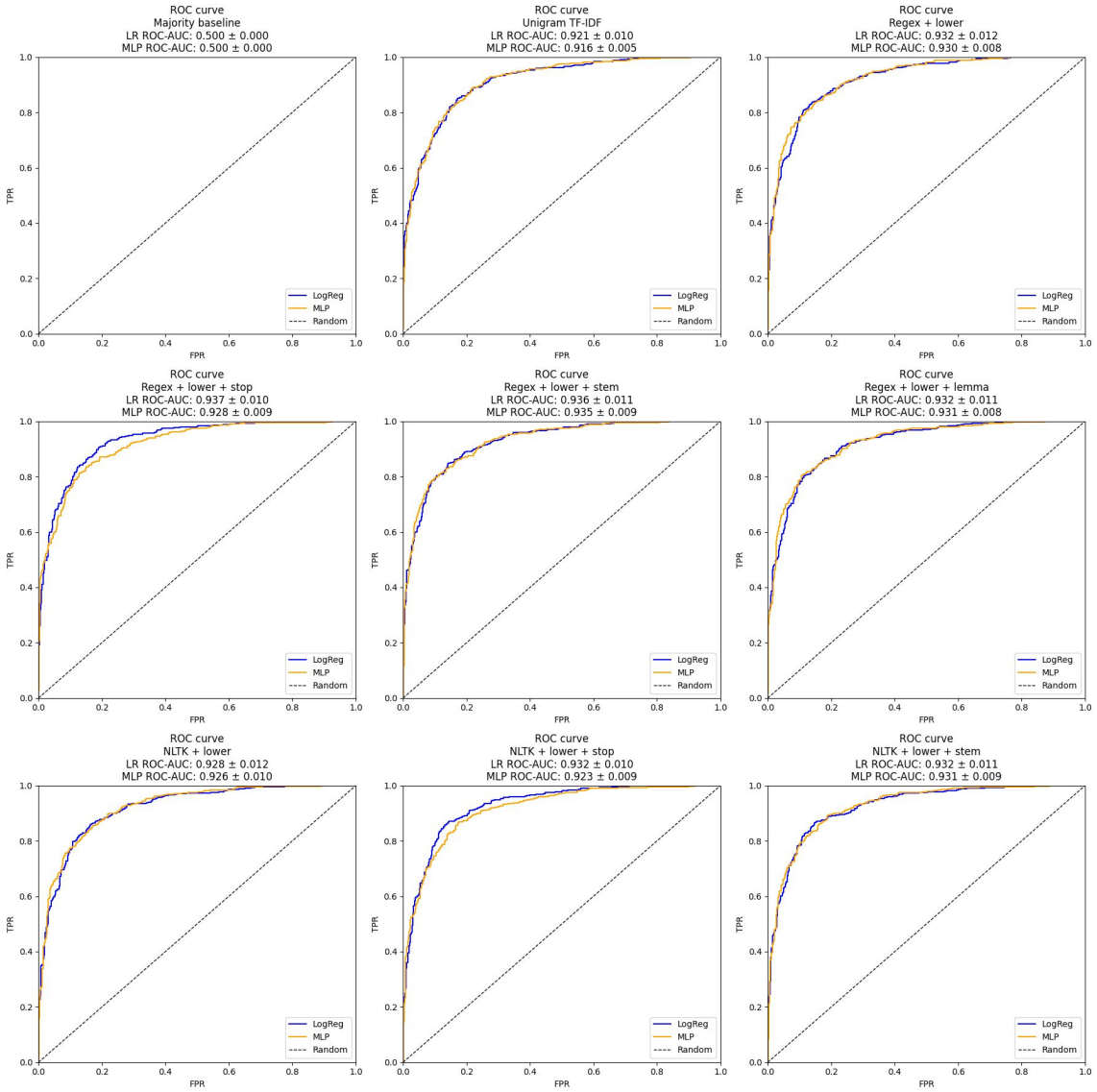
Configuration	Accuracy	F1 Score	ROC-AUC	PR-AUC
Majority baseline	0.4988 ± 0.0000	0.3328 ± 0.0000	0.5000 ± 0.0000	0.5013 ± 0.0000
Unigram TF-IDF	0.8344 ± 0.0065	0.8344 ± 0.0065	0.9163 ± 0.0045	0.9159 ± 0.0081
Regex + lower	0.8518 ± 0.0130	0.8518 ± 0.0130	0.9303 ± 0.0080	0.9293 ± 0.0081
Regex + lower + stop	0.8522 ± 0.0138	0.8522 ± 0.0138	0.9282 ± 0.0089	0.9277 ± 0.0082
Regex + lower + stem	$\mathbf{0.8544 \pm 0.0128}$	$\mathbf{0.8544 \pm 0.0128}$	$\mathbf{0.9349 \pm 0.0090}$	$\mathbf{0.9345 \pm 0.0095}$
Regex + lower + lemma	0.8488 ± 0.0099	0.8488 ± 0.0099	0.9305 ± 0.0079	0.9288 ± 0.0093
NLTK + lower	0.8466 ± 0.0154	0.8466 ± 0.0154	0.9262 ± 0.0097	0.9242 ± 0.0096
NLTK + lower + stop	0.8482 ± 0.0128	0.8482 ± 0.0128	0.9233 ± 0.0090	0.9204 ± 0.0111
NLTK + lower + stem	0.8520 ± 0.0134	0.8520 ± 0.0134	0.9307 ± 0.0087	0.9292 ± 0.0096

Results analysis









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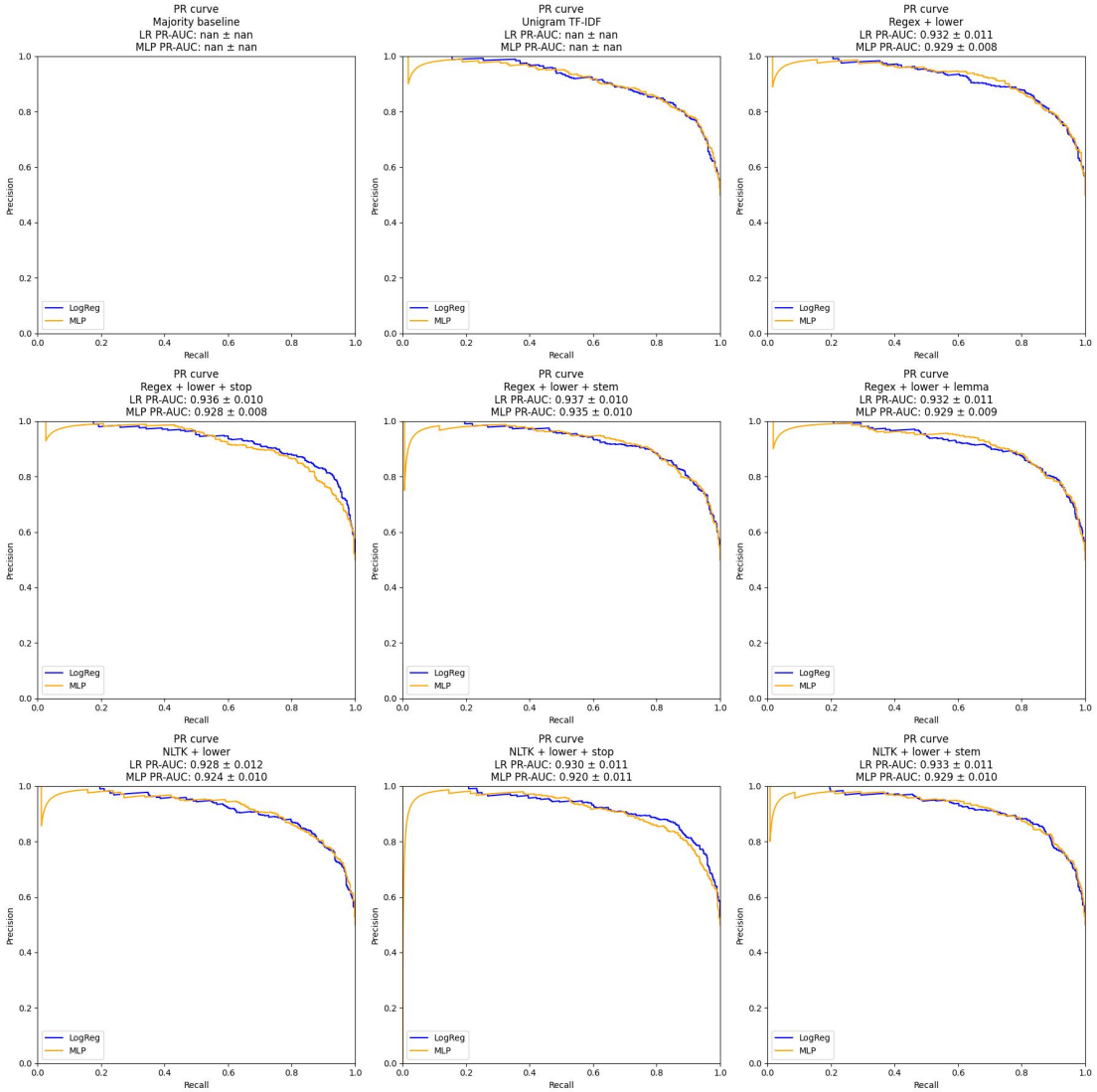

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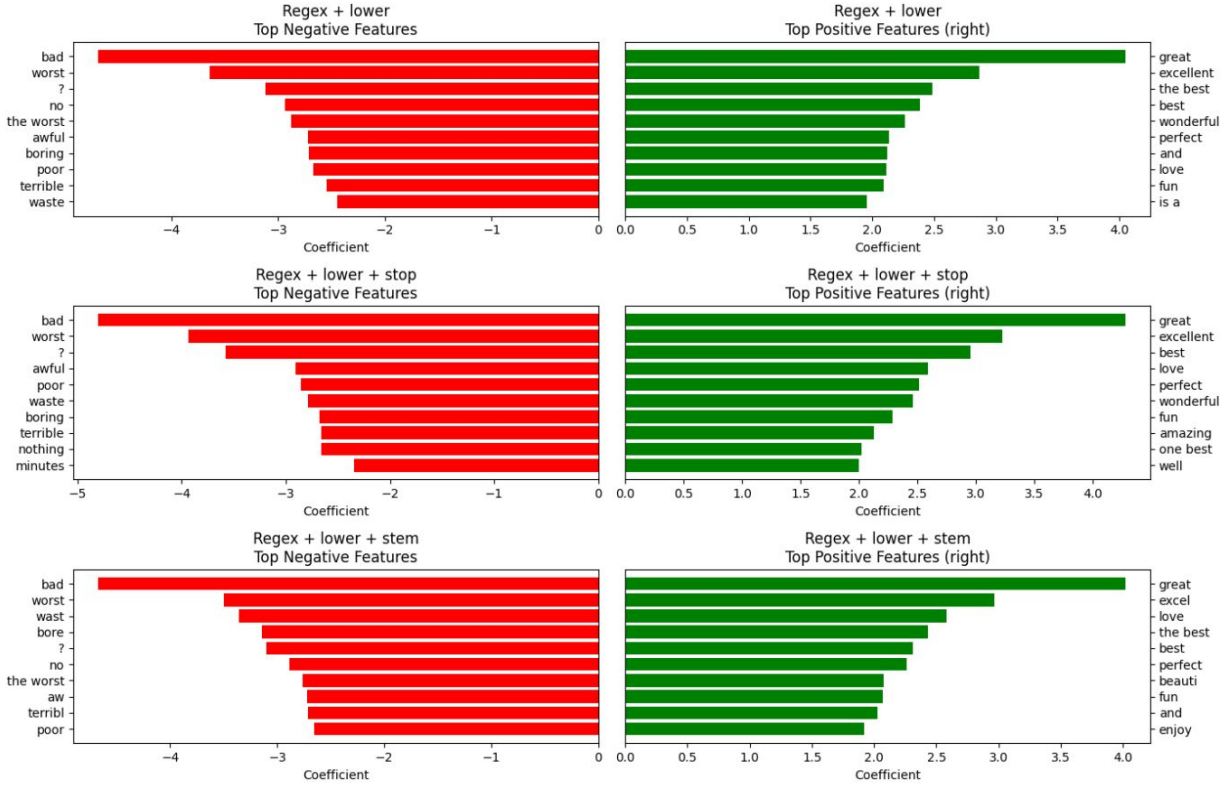

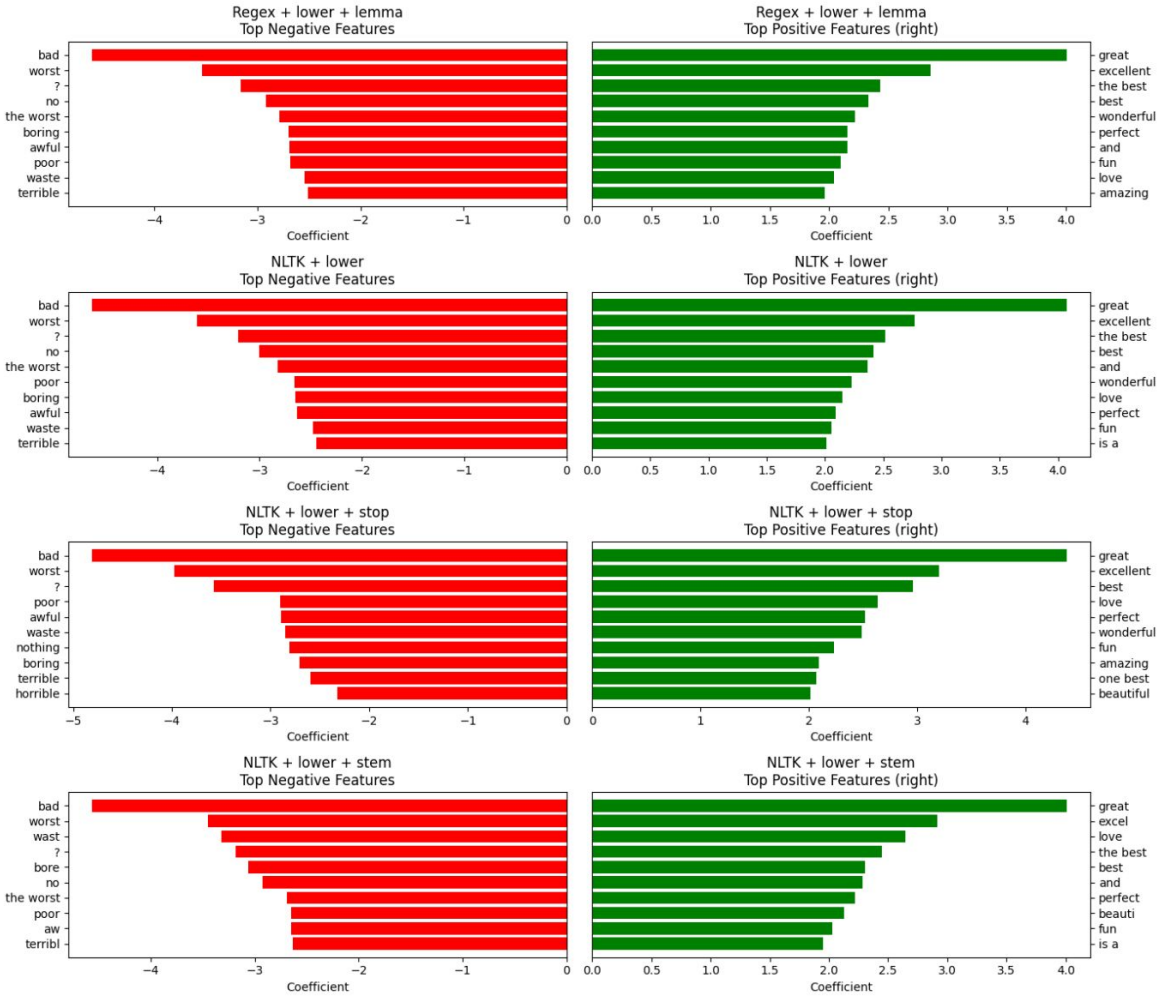

Conclusions and future directions

Classical NLP pipelines with TF-IDF and simple classifiers can achieve **strong results** on text classification tasks like **sentiment analysis**. The best model (Logistic Regression with regex tokenization, lowercasing, and stopword removal) achieved an F1 score of 0.8645 and ROC-AUC of 0.9375. **Careful preprocessing** and **model selection** are **important**, but more **complex pipelines** do not always guarantee better performance. The results show that even **simple linear models** can effectively capture sentiment patterns in movie reviews.

Future prospects include:

- **Error Analysis**: Examine misclassified reviews to identify common error patterns.
- **Threshold Tuning**: Explore different probability thresholds for classification.
- **Advanced Features**: Experiment with word embeddings (Word2Vec, GloVe) or transformer-based models (BERT) for comparison.

Thank you!

If you have any question, we're ready to answer them